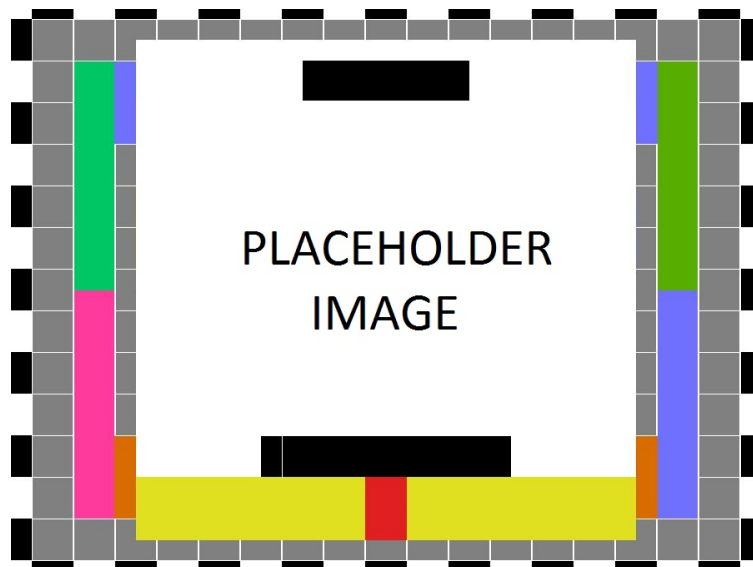


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**DETAILED DESIGN SPECIFICATION
CSE 4317: SENIOR DESIGN II
SPRING 2027**



**TEAM PERCEPTRON
DIGITDESK HANDWRITTEN FORM DIGITIZATION
SERVICE**

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REVISION HISTORY

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1.0	02.05.2027	AT, GH, CB	official release

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1 INTRODUCTION

Implementation-level design of DigitDesk Handwritten Form Digitization Service as an IEEE 1016-2009 [1] Software Design Description, building on the SRS and ADS (layers X/Y/Z).

2 DESIGN STAKEHOLDERS & CONCERNS

Implementers, integrators, testers, maintainers; concerns of composition, data, interfaces, algorithms, state, and resources.

3 DESIGN VIEWPOINTS

Context, Composition, Information, Interface, Interaction & State, Algorithm, and Resource viewpoints per IEEE 1016; irrelevant viewpoints omitted per subsystem.

4 SYSTEM OVERVIEW

Deployed as Docker containers on an on-prem Linux host with a message queue (RabbitMQ) between stages and PostgreSQL + object storage for persistence.

5 X LAYER SUBSYSTEMS — INGESTION & PREPROCESSING

5.1 LAYER RESOURCES

OS/Depends: Linux, Python, OpenCV, Pillow; **Language:** Python.

5.2 X.SEGMENT

Composition: a worker consuming preprocessed pages. **Information:** Field{form_id, field_id, bbox, crop[]}. **Interface (#01 chain):** in preprocessed page; out field crops to queue. **Algorithm:** affine-align the page to the template via fiducial marks, then crop fixed field boxes; reject pages whose alignment residual exceeds a threshold. **Rationale:** deterministic and testable for a fixed template (AD-01).

6 Y LAYER SUBSYSTEMS — RECOGNITION & ANALYTICS

6.1 LAYER RESOURCES

Depends: PyTorch, NumPy; **Language:** Python; GPU optional.

6.2 Y.RECOGNIZER

Information: input Field; output Prediction{value, logits[], model_version, data_hash} (MS-01). **Interface (#02):** in field crops; out predictions. **Algorithm:** a small CNN (LeNet/MobileNet-class) over per-digit crops; the field value is the ordered concatenation of per-digit argmax. **Resource:** model artifact loaded from a versioned path (QA-01).

6.3 Y.CALIBRATION & Y.ROUTER

Y.Calibration: temperature-scaled softmax fit on a held-out set converts logits to a calibrated field confidence. **Y.Router:** if $\text{conf} \geq \text{threshold}$: accept else: enqueue_review (FR-03); threshold in config.

7 Z LAYER SUBSYSTEMS — SERVICE, STORAGE & REVIEW

7.1 LAYER RESOURCES

Depends: FastAPI, PostgreSQL, React (review UI); **Languages:** Python, TypeScript.

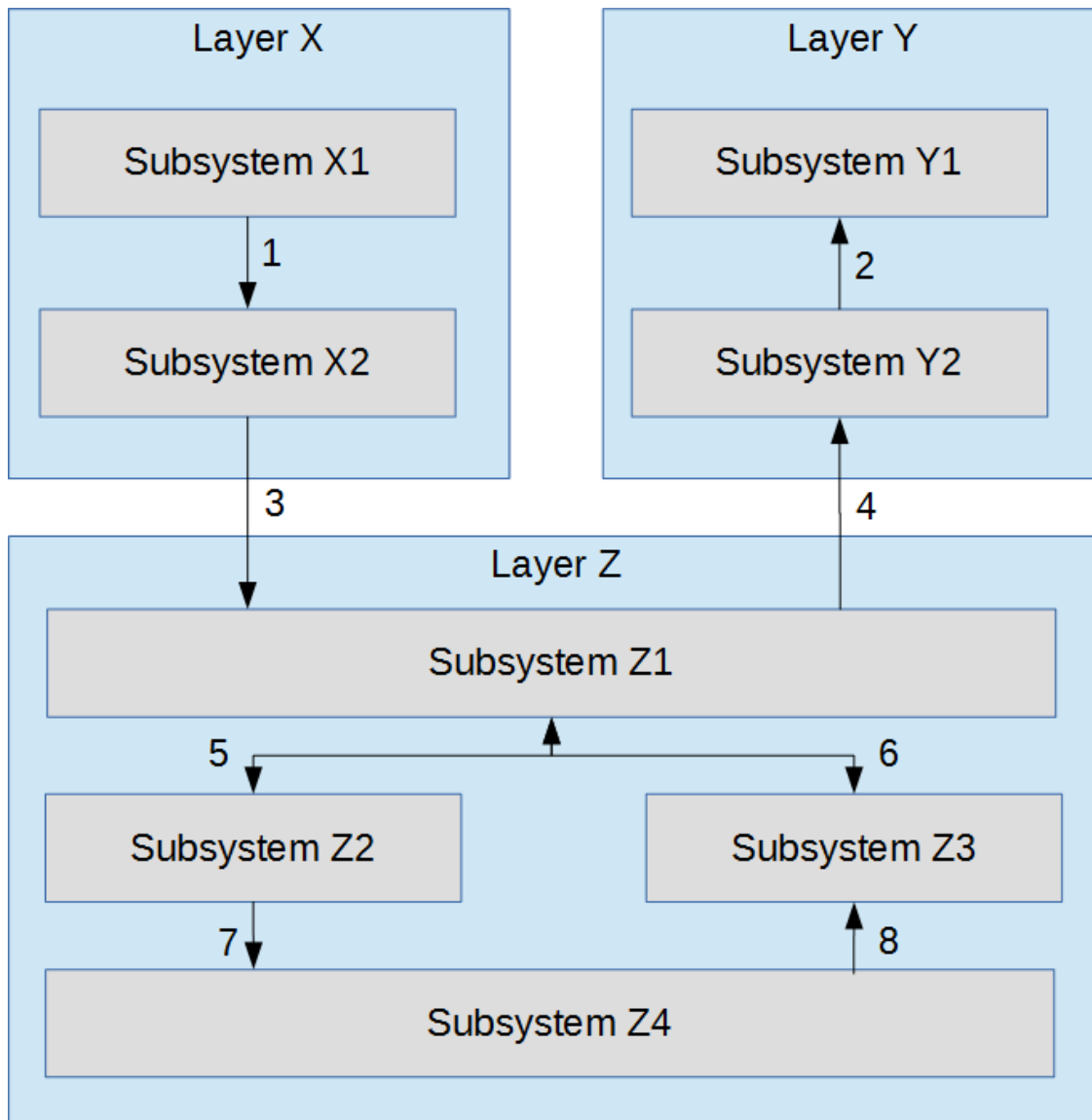


Figure 1: Pipeline carried forward from the ADS

7.2 Z.API & Z.STORE

Interface: POST /forms (EI-01), GET /jobs/{id}, GET /export. **Information:** Record{form_id, field_id, value, source:auto/reviewed, model_version, reviewer_id?} in PostgreSQL; images in on-prem object storage (SE-01). Authentication via the on-prem IdP (SE-02).

7.3 Z.REVIEWUI & Z.AUDIT

Z.ReviewUI: React screen showing crop + proposed value with keyboard confirm/correct (FR-04, US-01). **Z.Audit:** append-only audit table; every decision writes {who, when, field, old, new} (FR-05).

8 REQUIREMENTS TRACEABILITY

Req	Requirement (abbrev.)	Component	Iface
FR-01	Field segmentation	X.Segment	#01
FR-02	Recognition+conf	Y.Recognizer, Y.Calibration	#02
FR-03	Routing	Y.Router	—
FR-04	Review	Z.ReviewUI	—
FR-05	Export/audit	Z.API, Z.Audit	#exp
MS-01	Model versioning	Y.Recognizer, Z.Store	—

9 APPENDIX A

Database schema, API specification (OpenAPI), and the model card are attached to the closeout package.

REFERENCES

- [1] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.